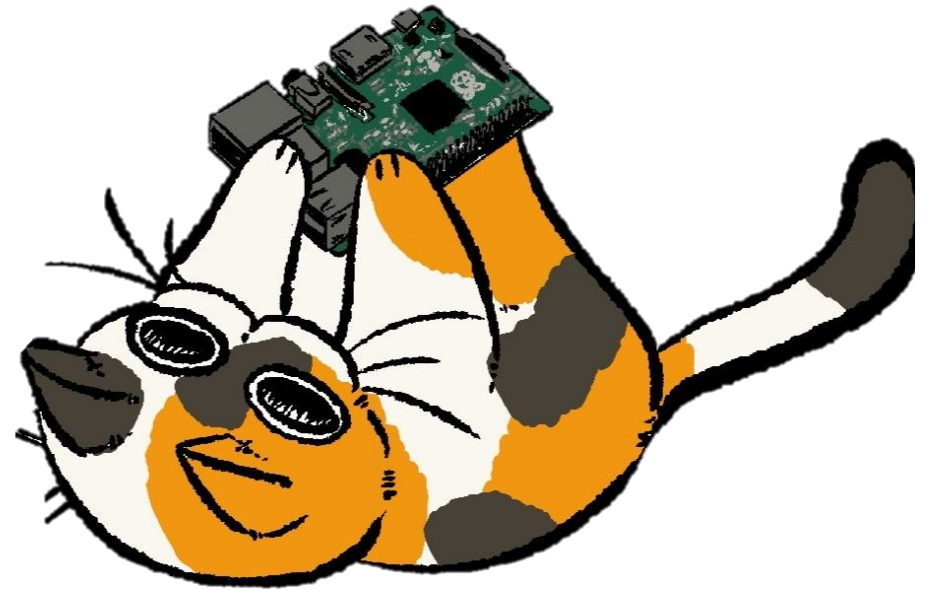




Student Branch Chapter @ USF

NVIDIA GPUs and AI Hardware
Hardware Lead: Liam Osman

Check in! Support Us



Follow Us



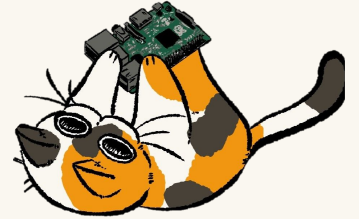
Discord

LinkedIn

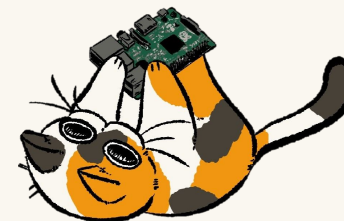
Instagram

Bulls
Connect

GitHub



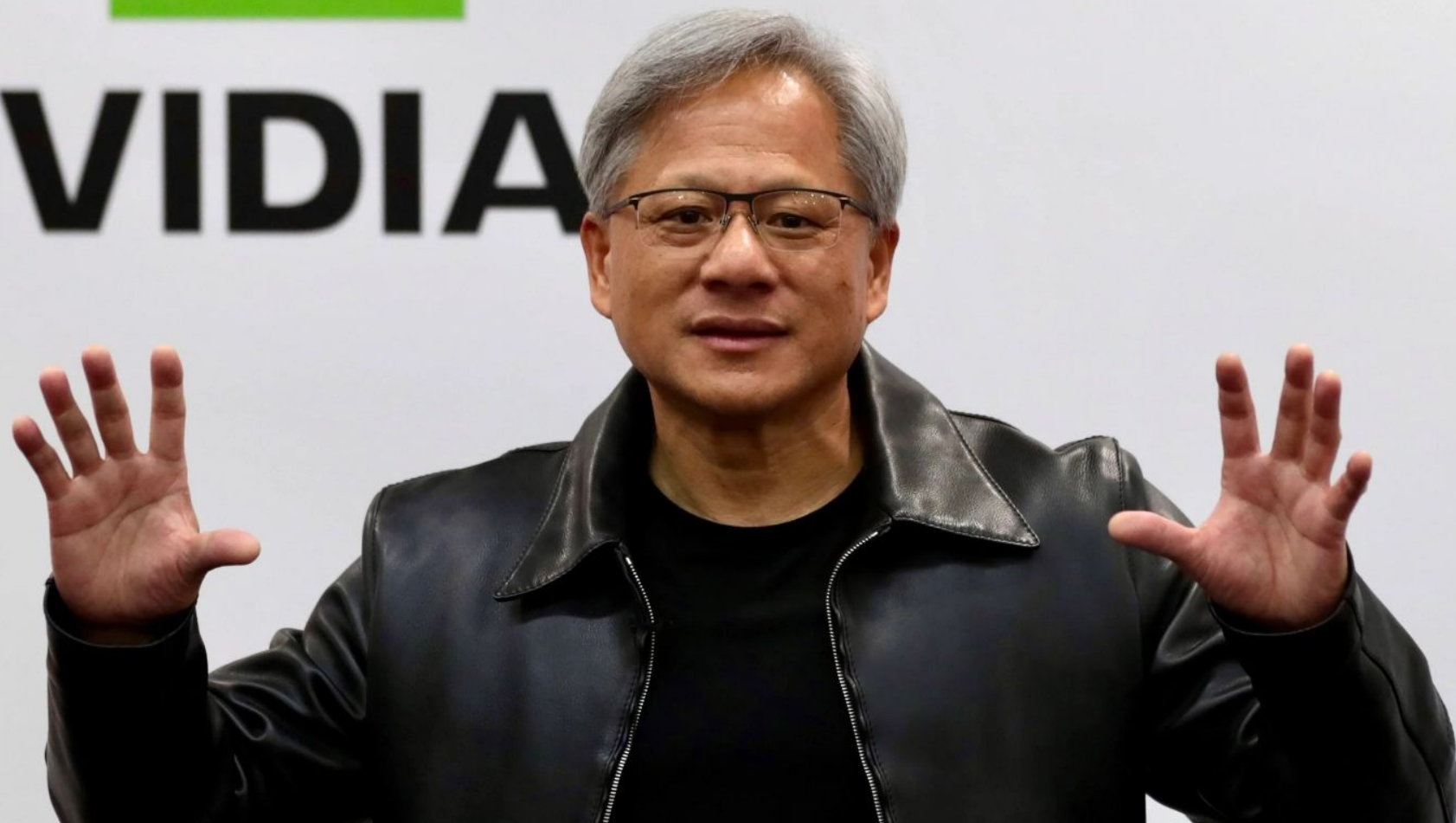
What are we doing today?



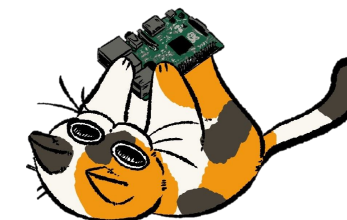
- **NVIDIA**
- **AI Training**
- **What's in a GPU anyways**
- **New Features**



NVIDIA



NVIDIA *(Guess When ChatGPT Came Out)*

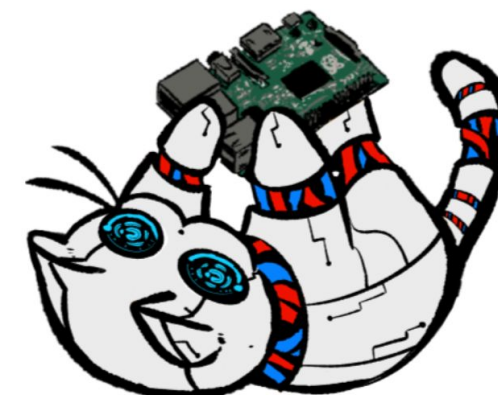
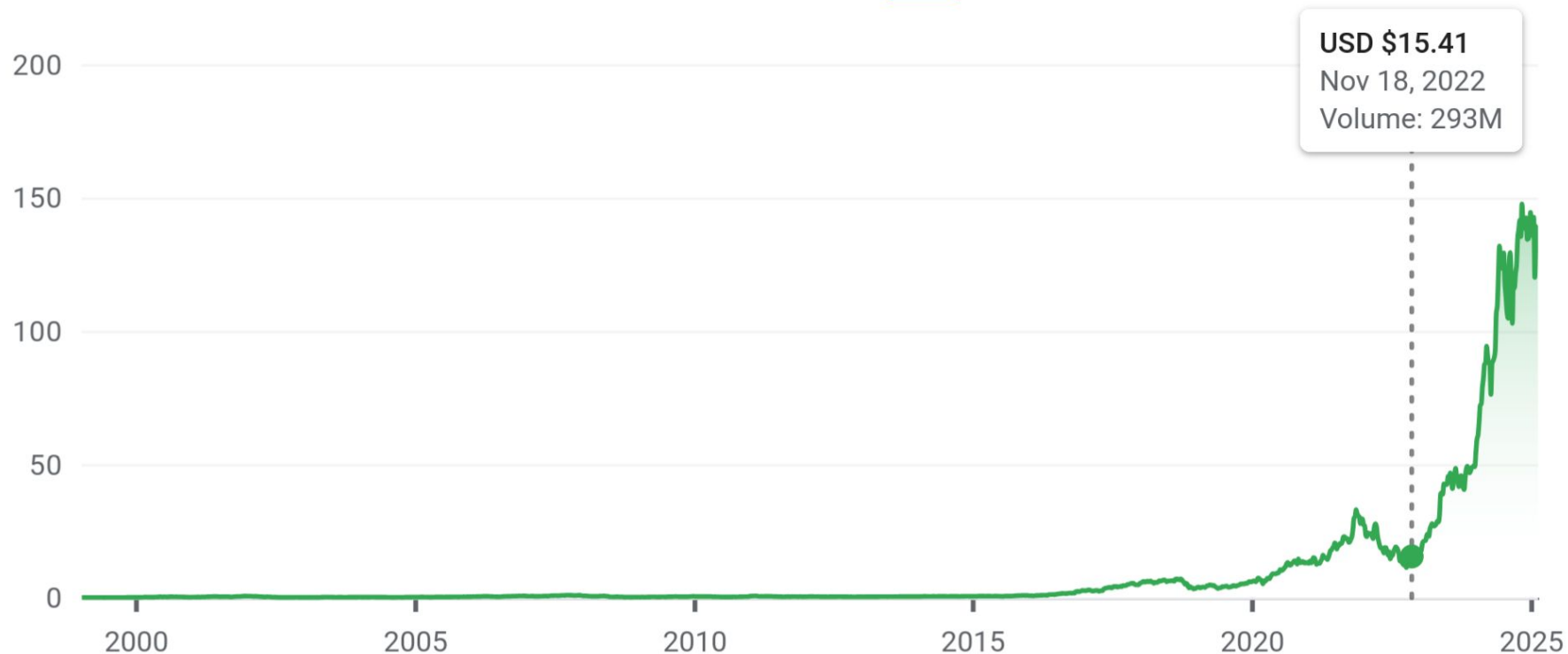


\$139.40 ↑ 348,400.00% +139.36 MAX

After Hours: \$139.10 (↓ 0.22%) -0.30

Closed: Feb 18, 7:59:58 PM UTC-5 · USD · NASDAQ · Disclaimer

1D 5D 1M 6M YTD 1Y 5Y MAX



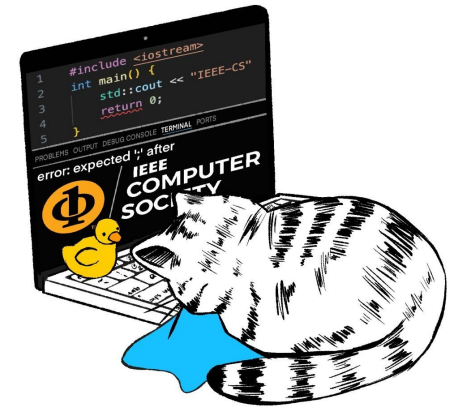
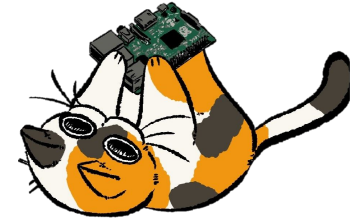
AMD *(Market Cap 18.5x Less Than NVDA)*

\$114.28 ↑ 595.13% +97.84 MAX

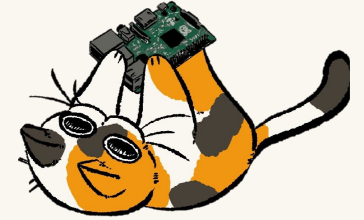
After Hours: \$114.45 (↑ 0.15%) +0.17

Closed: Feb 18, 8:00:00 PM UTC-5 · USD · NASDAQ · Disclaimer

1D 5D 1M 6M YTD 1Y 5Y MAX

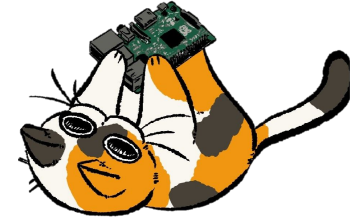


What is NVIDIA Doing?



- **NVIDIA has more employees than AMD, despite not making CPUs!**
- **AMD GPUs often have higher raw performance than NVIDIA ones**
- **Asides playing video games, what have they all been doing?**

Historical NVIDIA Software

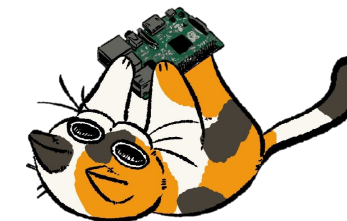


 **NVIDIA[®]**
PHYSX[®]

 **NVIDIA**
G-SYNC[™]

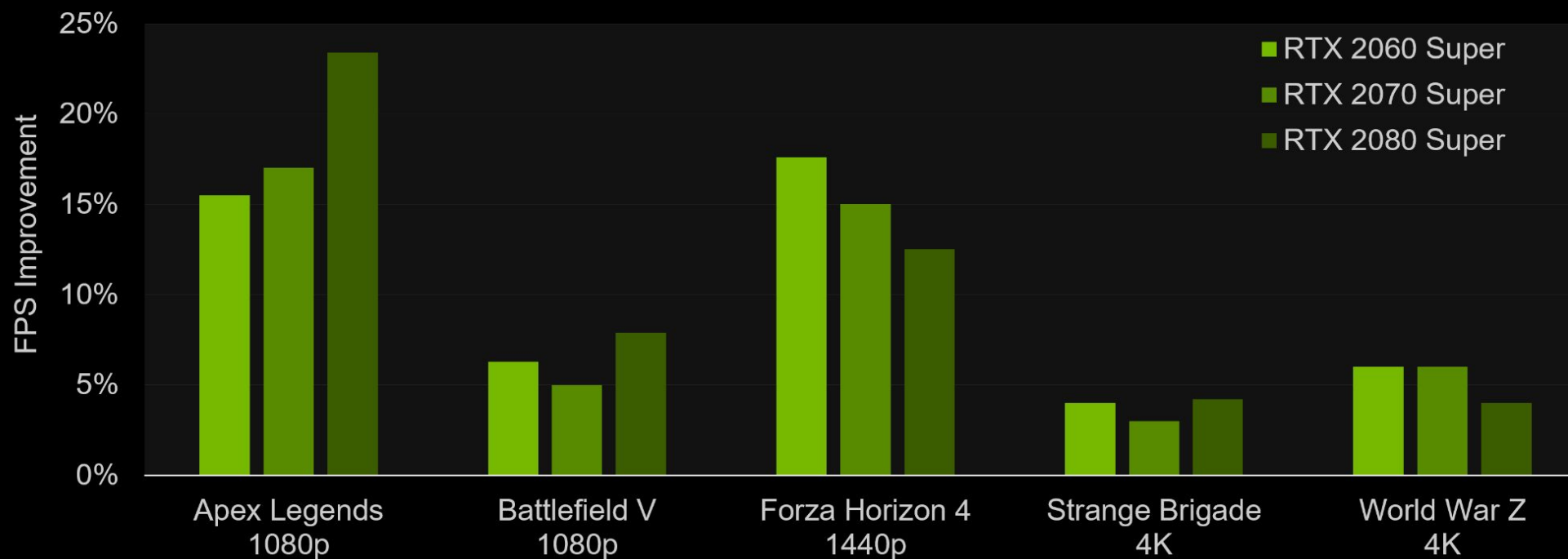
 **GEFORCE[®]**
EXPERIENCE[™]

Drivers



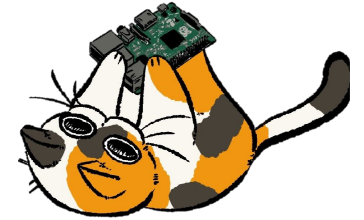
UP TO 23% FASTER PERFORMANCE

GAMESCOM GAME READY DRIVER



*Test System: Intel Core i9-7900X 3.3GHz, 16GB Corsair DDR4, Win 10 RS5 x64, performance gain is R435 over current R431.60 driver

The Importance of CUDA Cores



- **CUDA cores are NVIDIA's name for processing units on a GPU chip**
- **Does highly parallelized floating point (fractional) math operations**
- **Supported and used everywhere with software optimizations:**
 - **Data science and scientific calculations**
 - **Cryptography and breaking encryption**
 - **Blockchain hashing / cryptocurrency mining**

GeForce RTX 4090

GeForce RTX 4080 (16GB)

GeForce RTX 4080 (12GB)

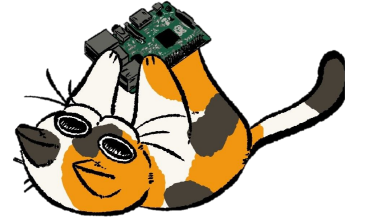
NVIDIA CUDA Cores

16384

9728

7680

Workstation Tailored Cards

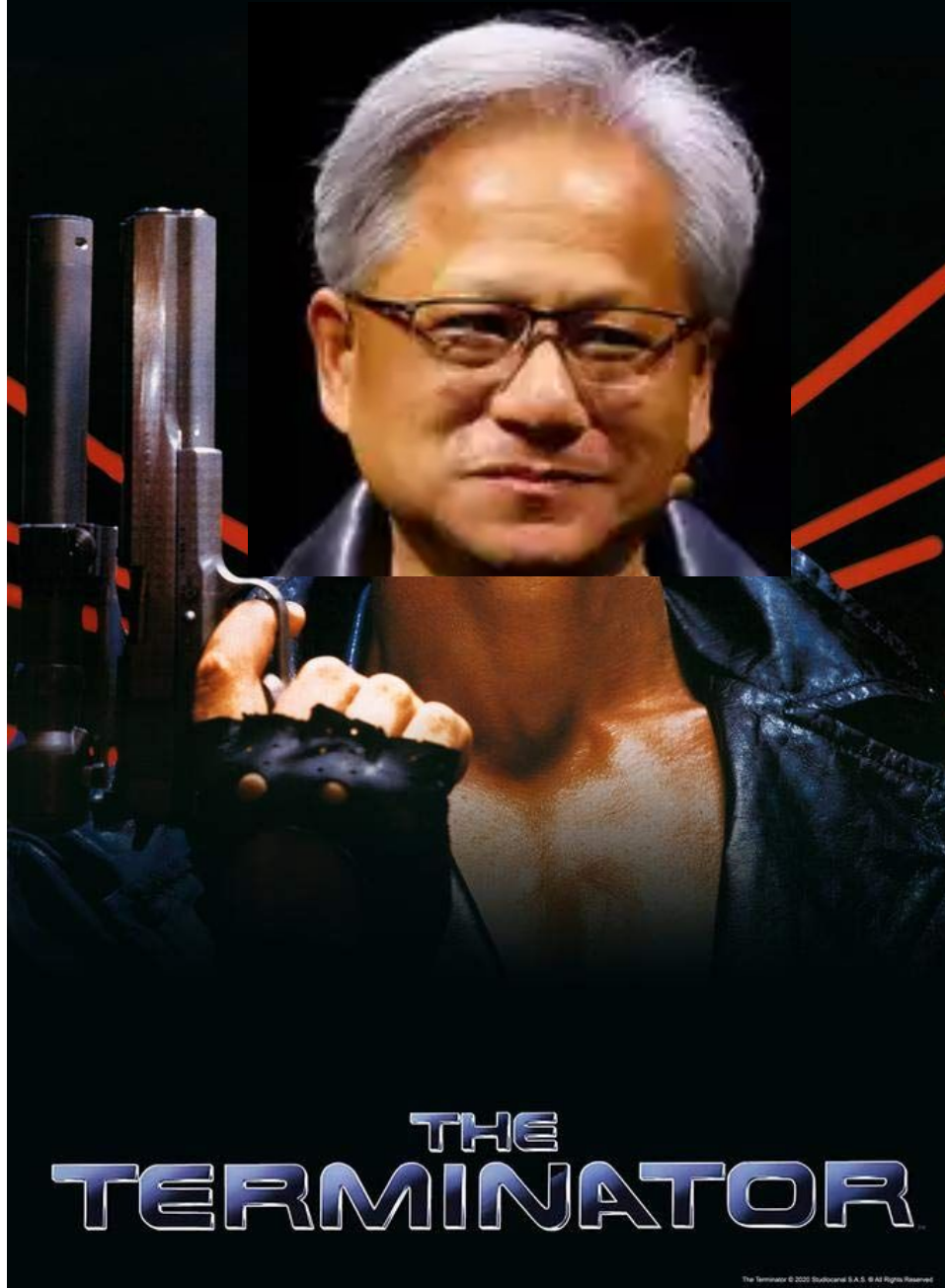
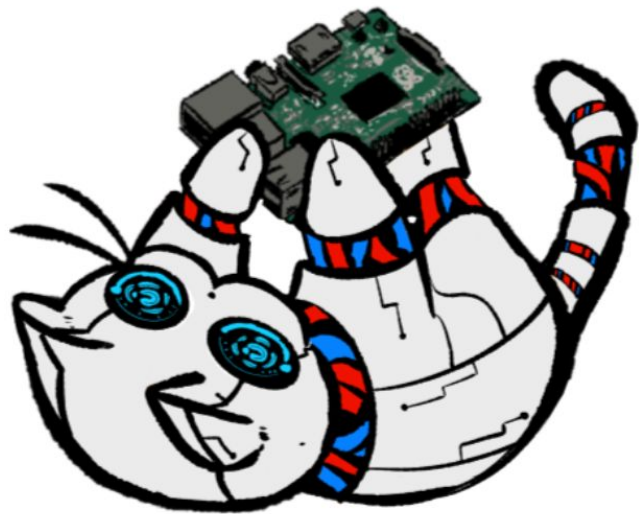


Pivot Away From Gamers

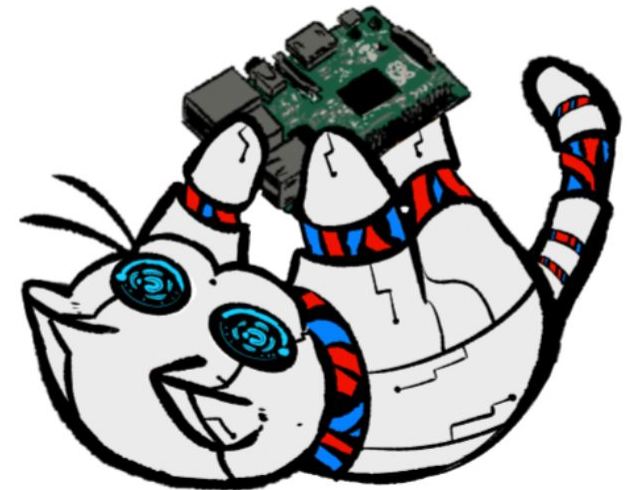
- There's a limited number of gamers in the world
- Gamers aren't shelling out infinite money
- Jensen Huang needs more leather jackets
- Why bother making video cards for gamers?



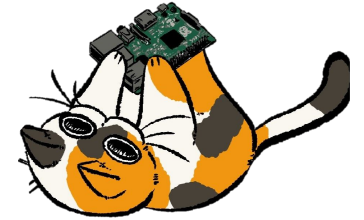
DEEP



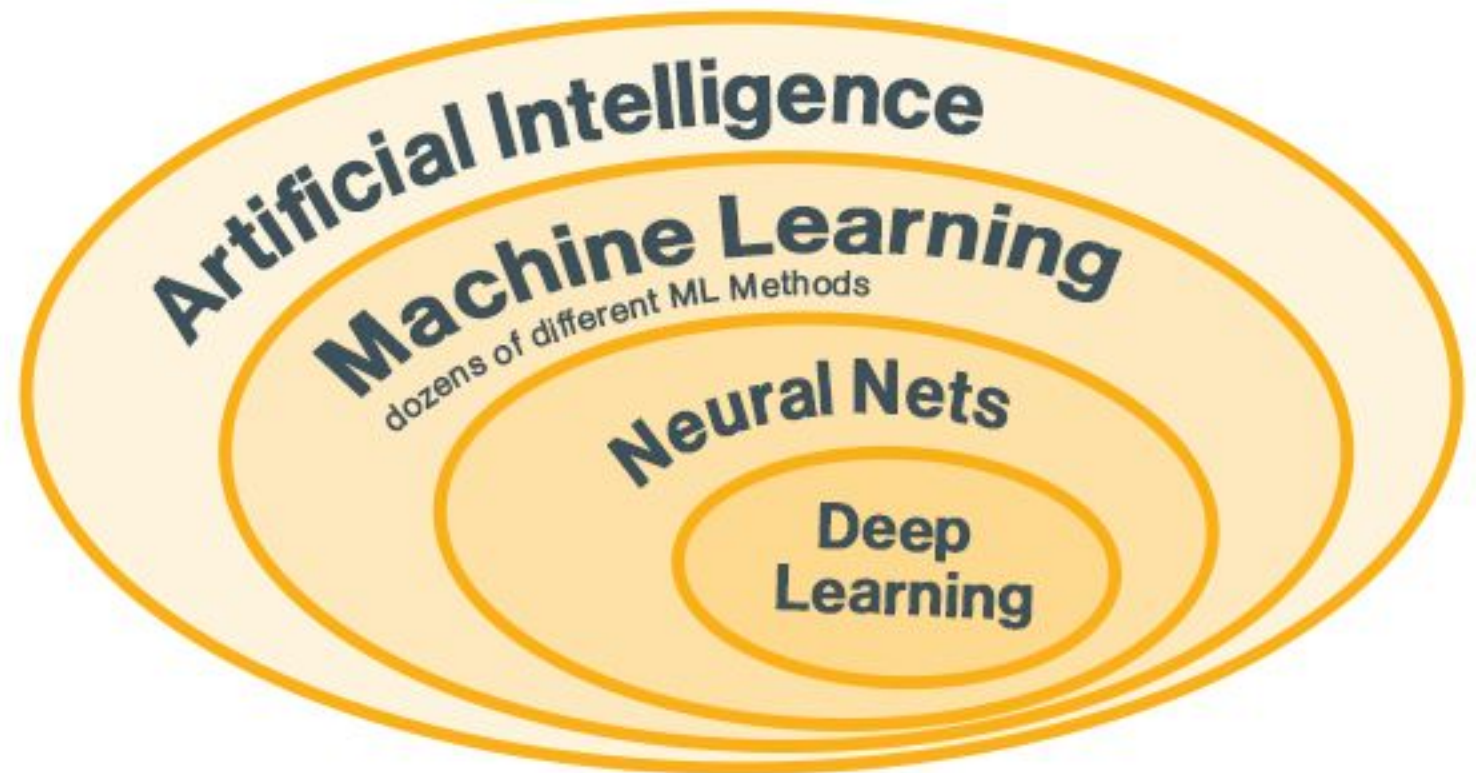
LEARNING



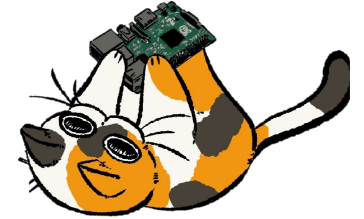
I'm Afraid I Can't Do That



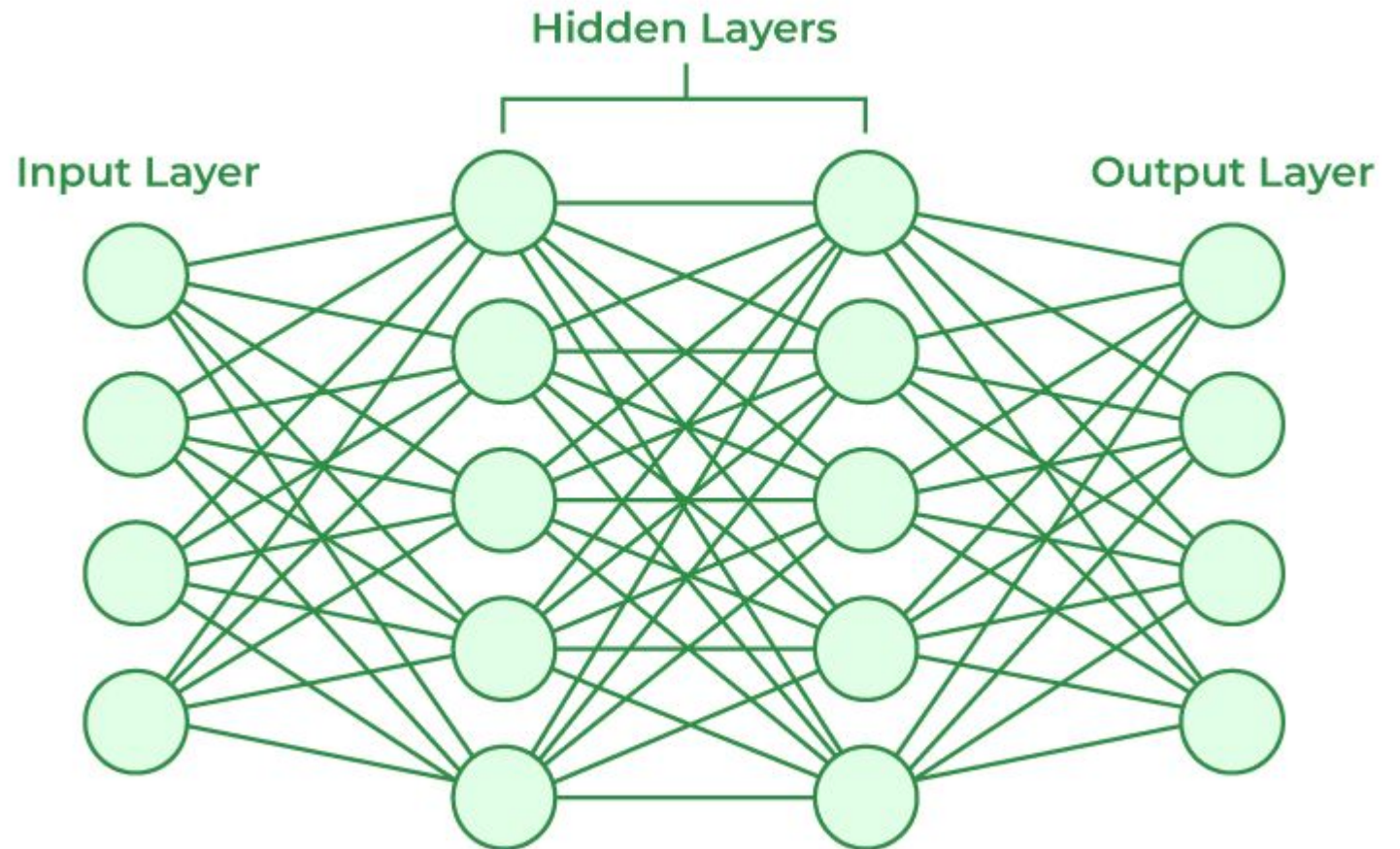
- AI investment is an infinite money glitch
- Skynet would be cool?



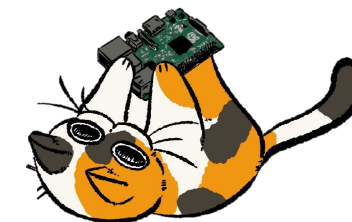
Deep Learning



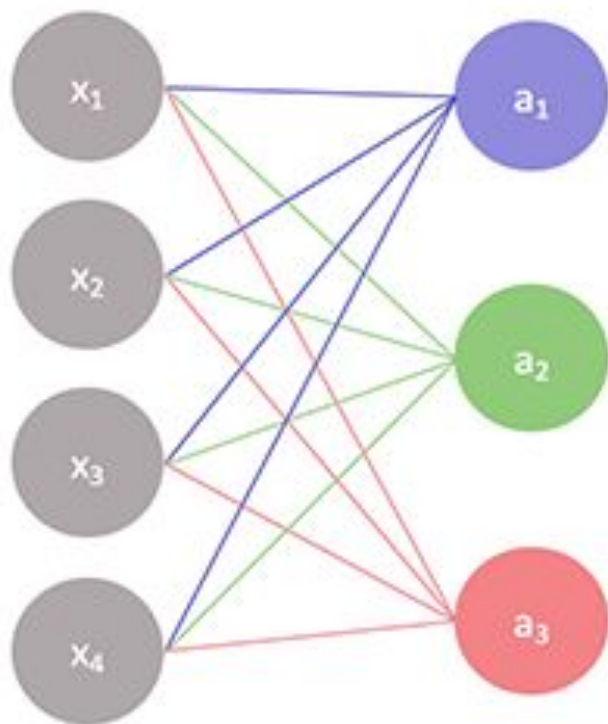
- Used in ChatGPT, Dall-E
- Very “Smart”
- Features many nodes
- Takes a lot of data



How To Train Your LLM

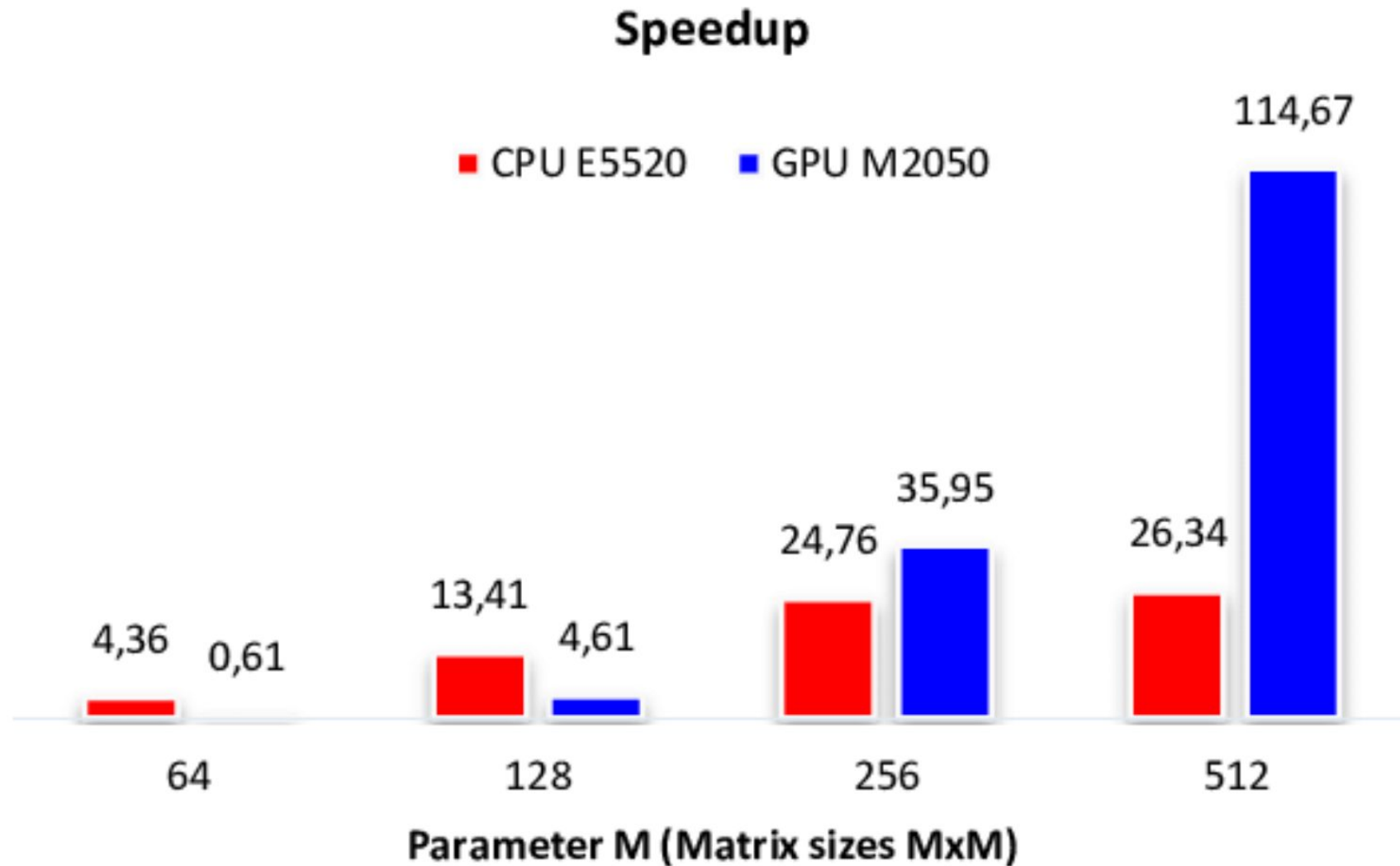
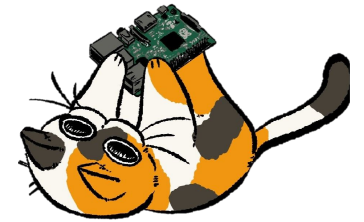


- Linear Algebra and other nerdy math stuff



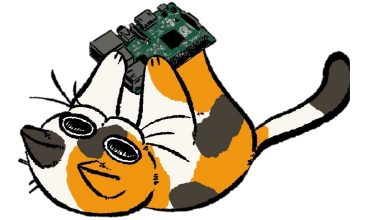
$$\begin{bmatrix} w_1 & w_2 & w_3 & w_4 \\ w_1 & w_2 & w_3 & w_4 \\ w_1 & w_2 & w_3 & w_4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} b \\ b \\ b \end{bmatrix} = \begin{bmatrix} w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4 + b \\ w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4 + b \\ w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4 + b \end{bmatrix} \rightarrow \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$$

Performance Diff

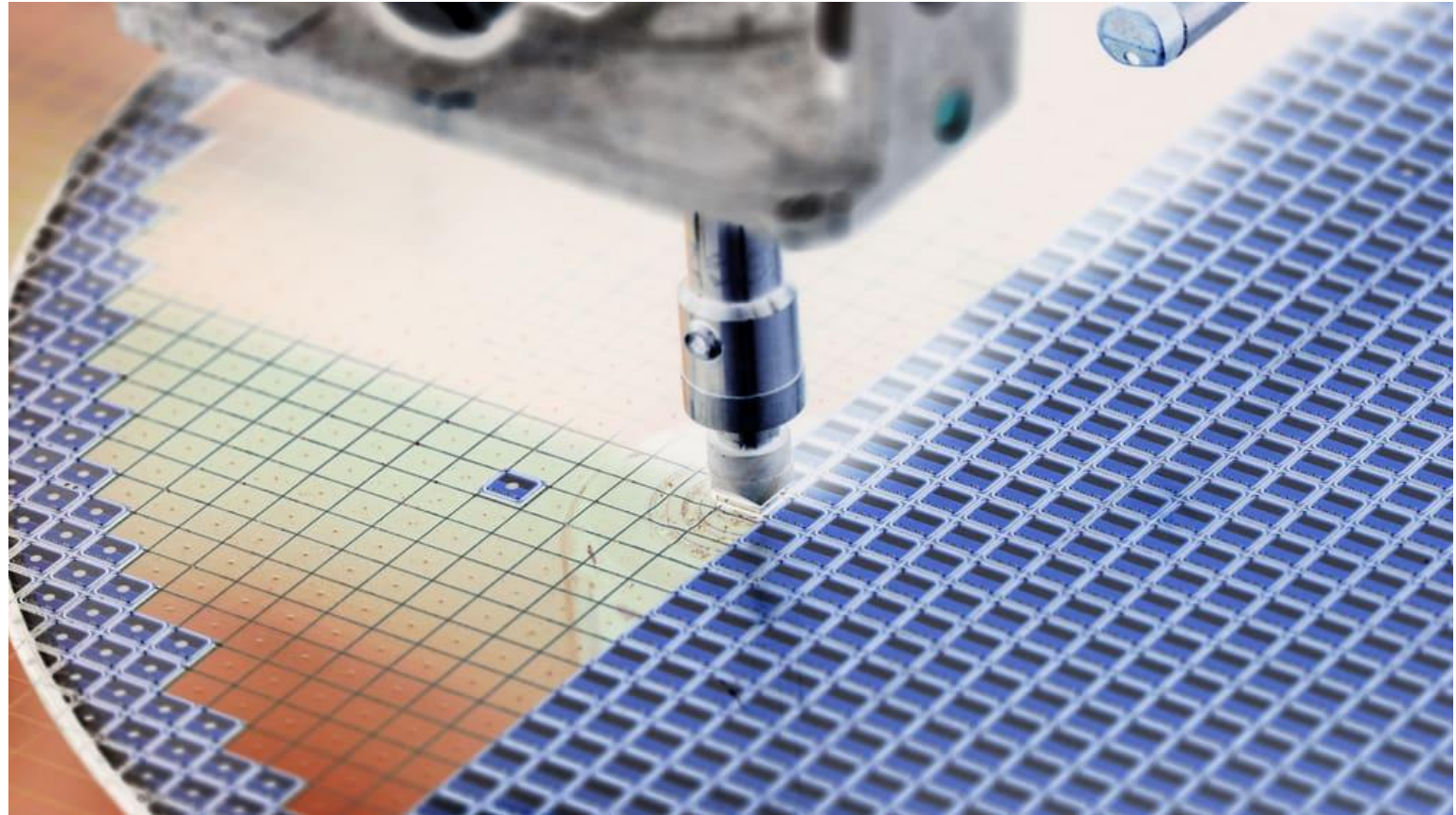




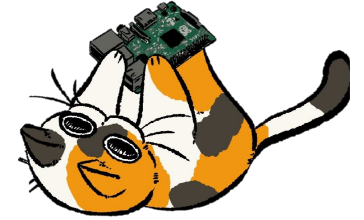
Why So Expensive?



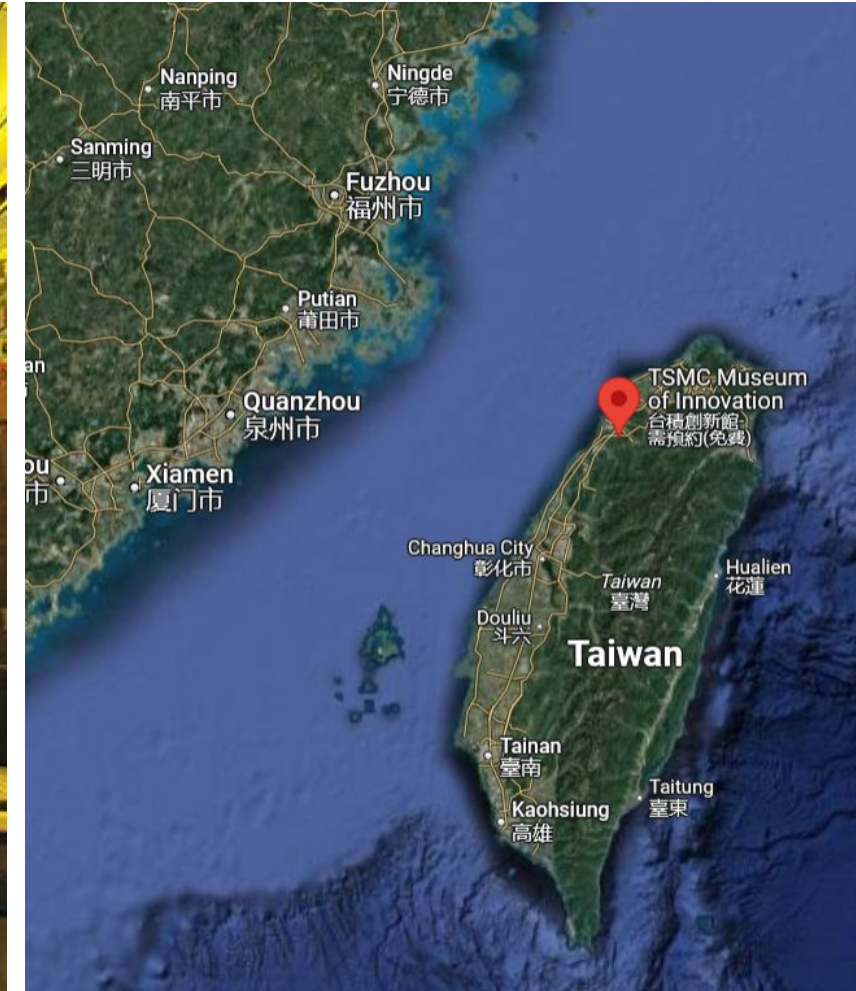
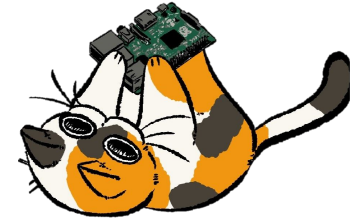
- **Very precise process**
- **Lots of monopolies**
- **Very valuable**
- **You'll pay**

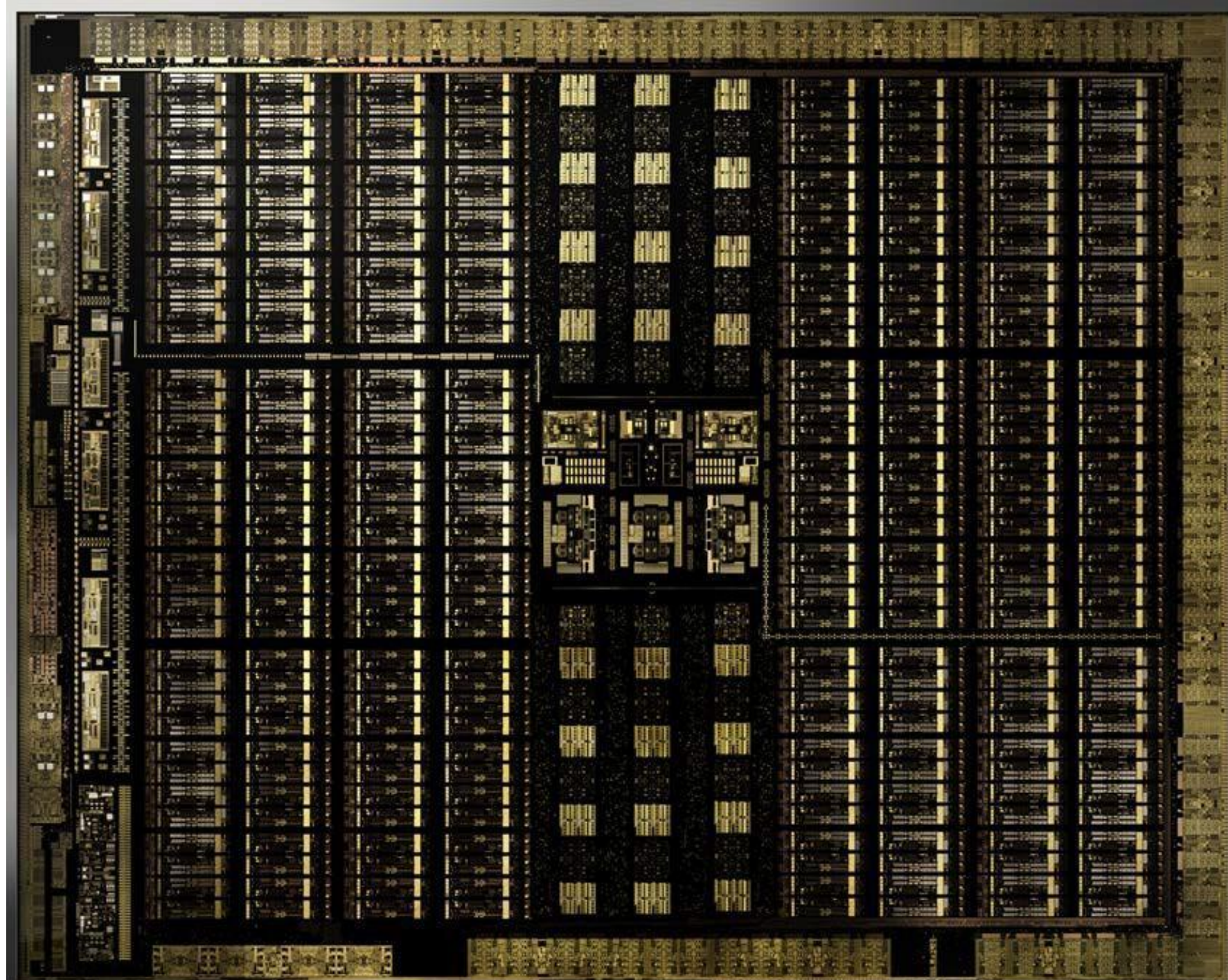


EUV Lithography - \$380M Each



Manufacturing - \$∞

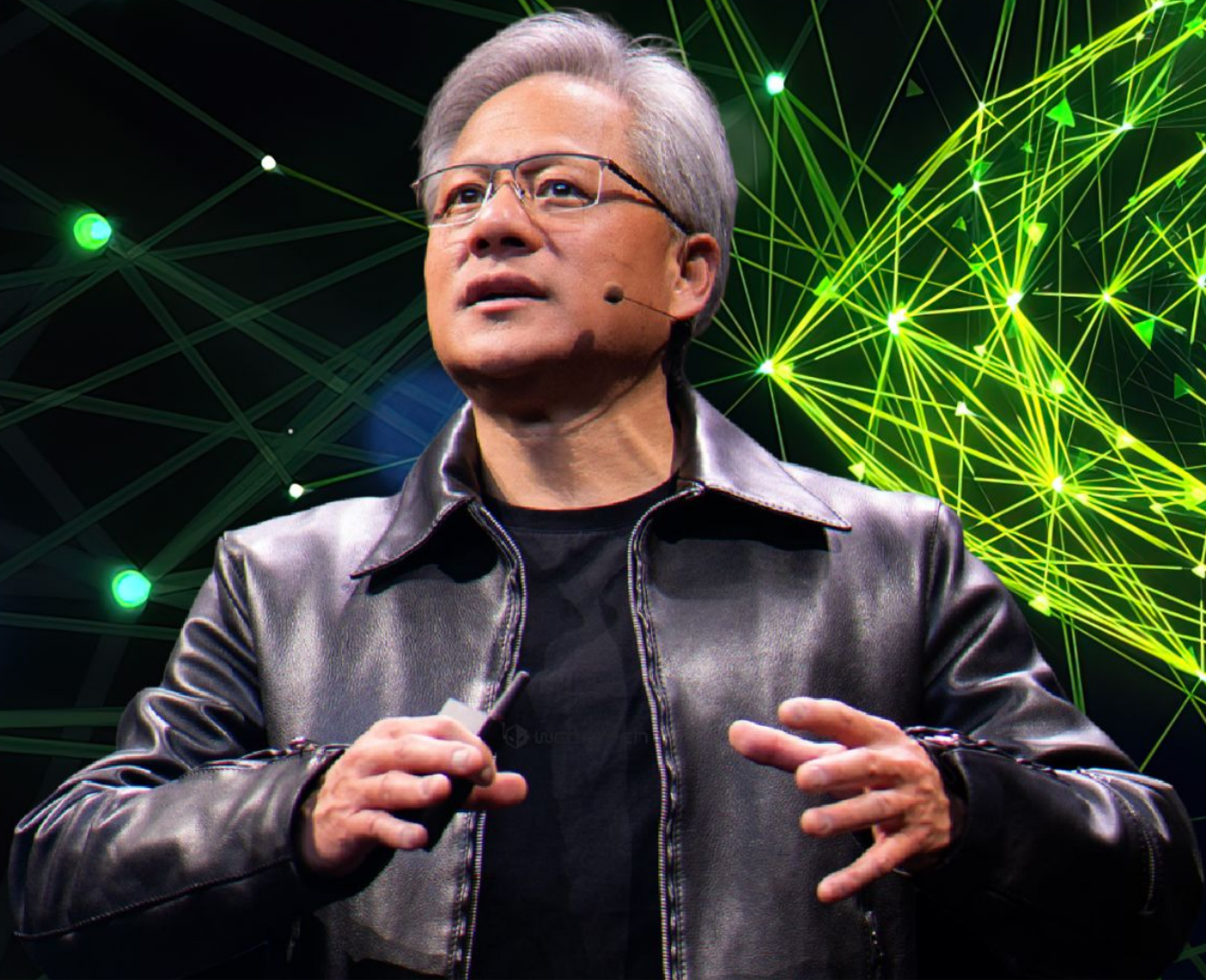




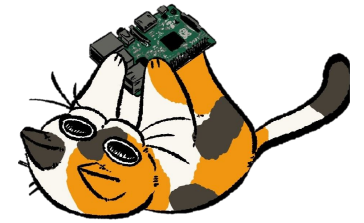
Die Hardware

- Regular INT/FP CUDA cores
- Tensor Cores built just to do matrix math
- RT Cores built to do Ray Tracing
- Memory and other boring stuff

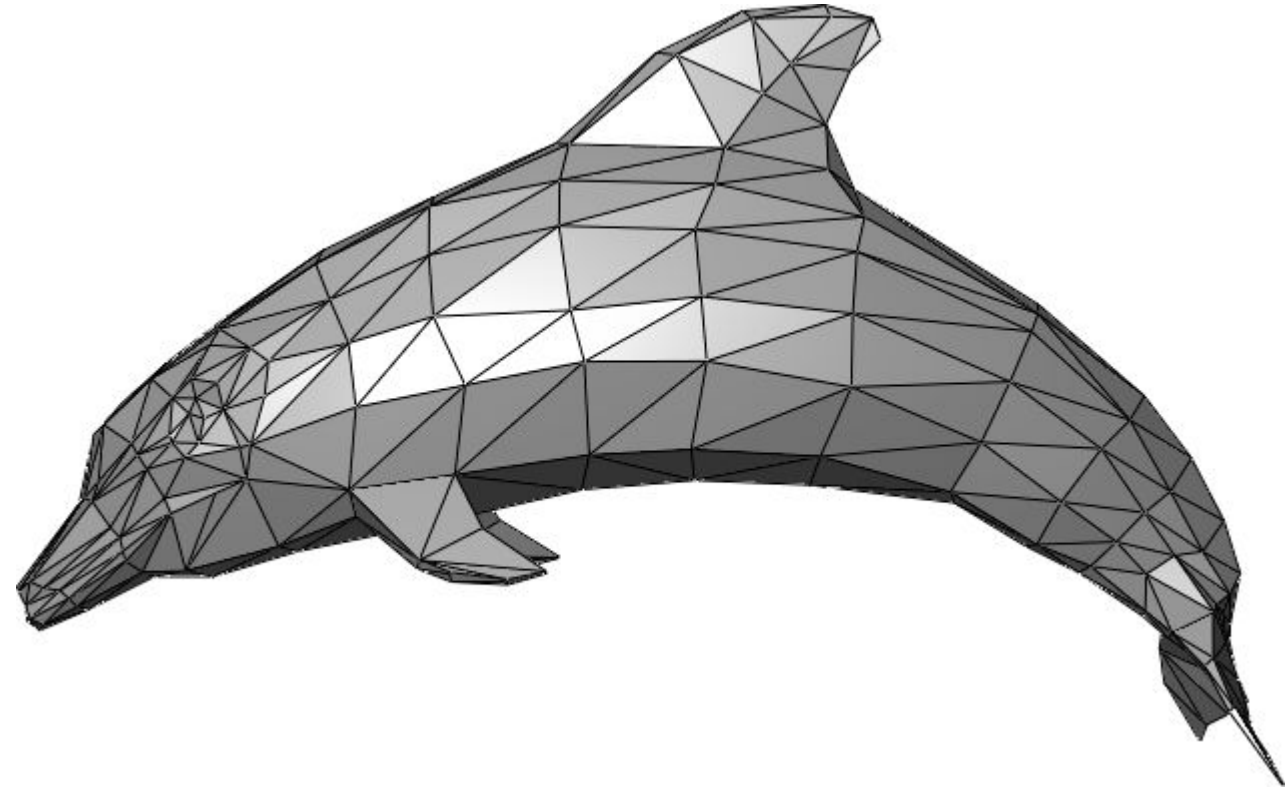
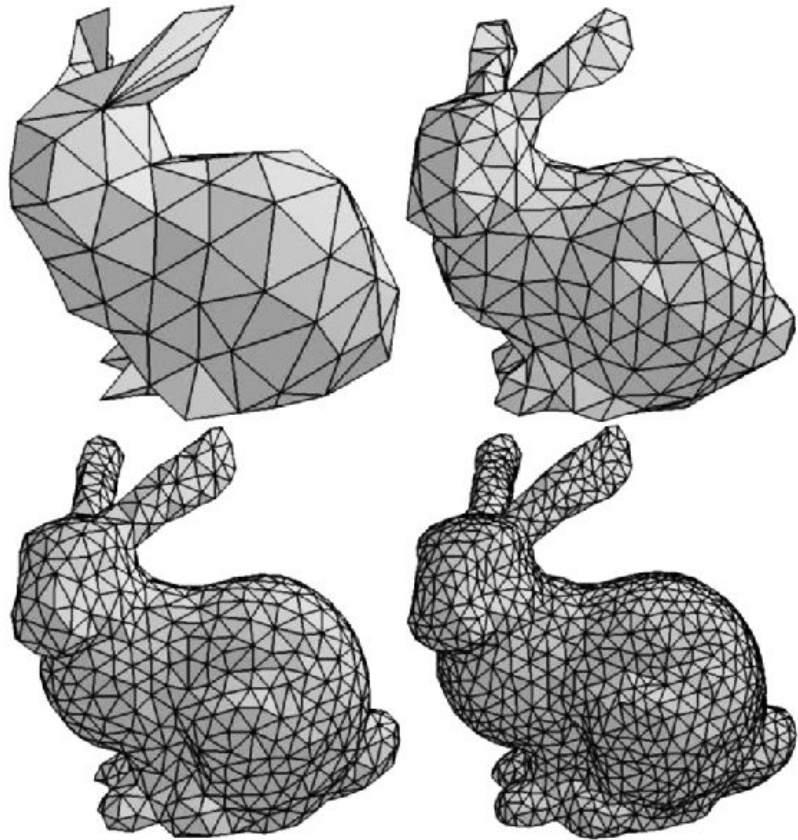




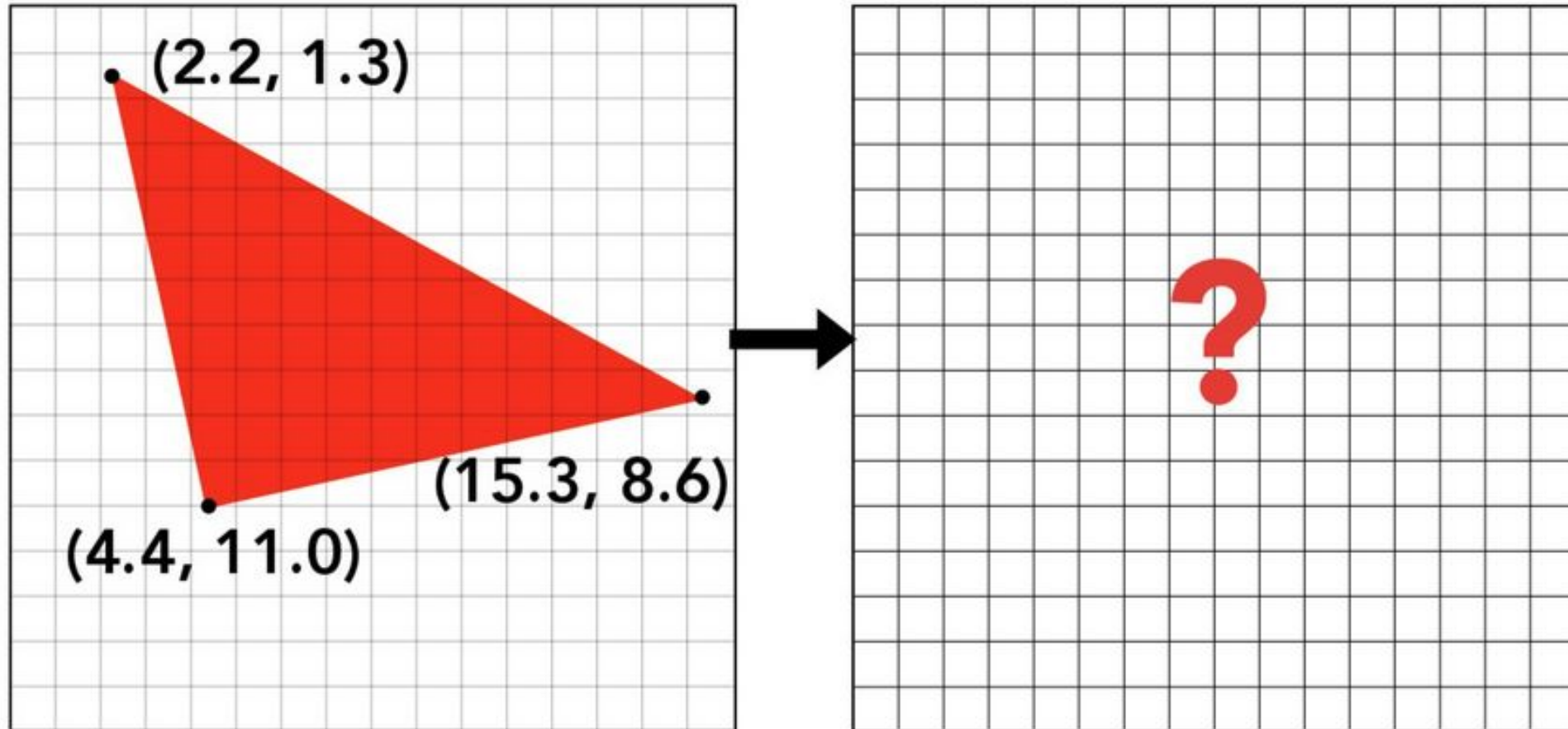
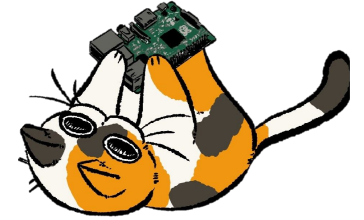
3D Modelling



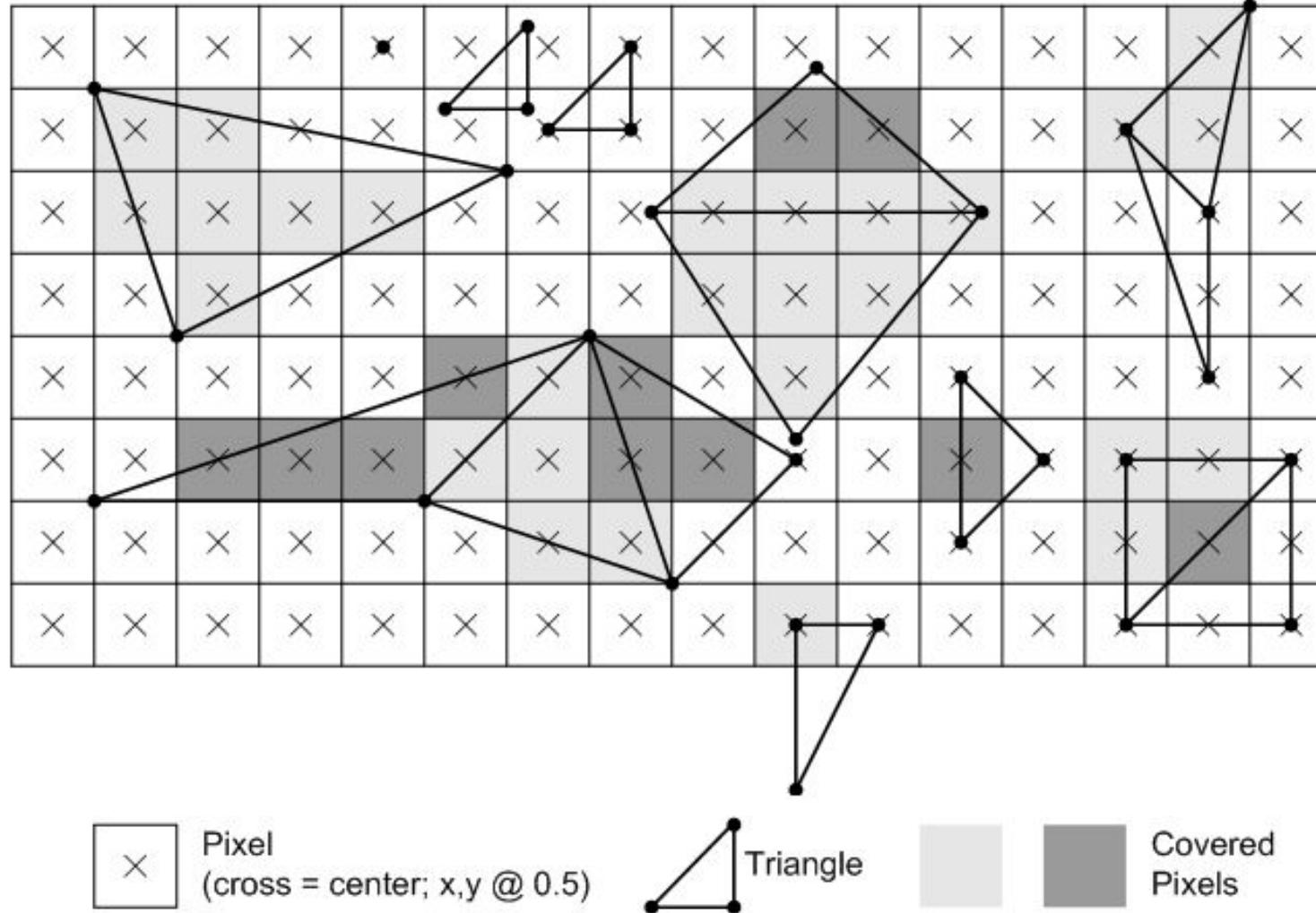
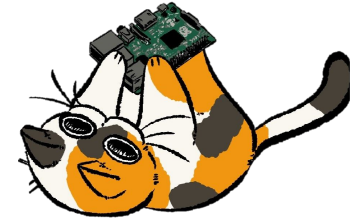
It's all triangles!



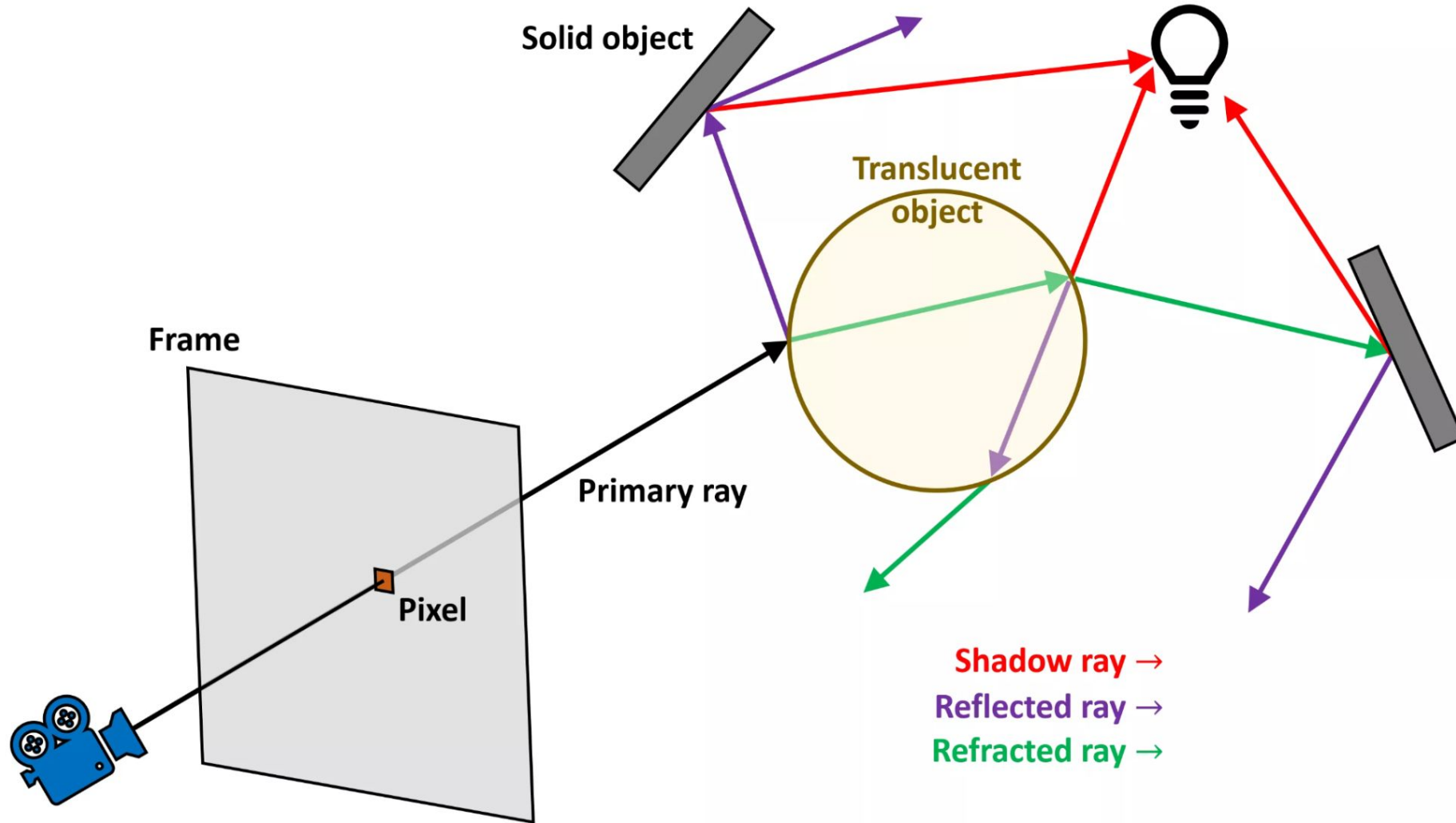
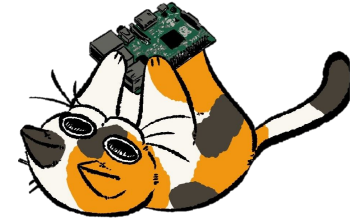
Rasterization



Rasterization



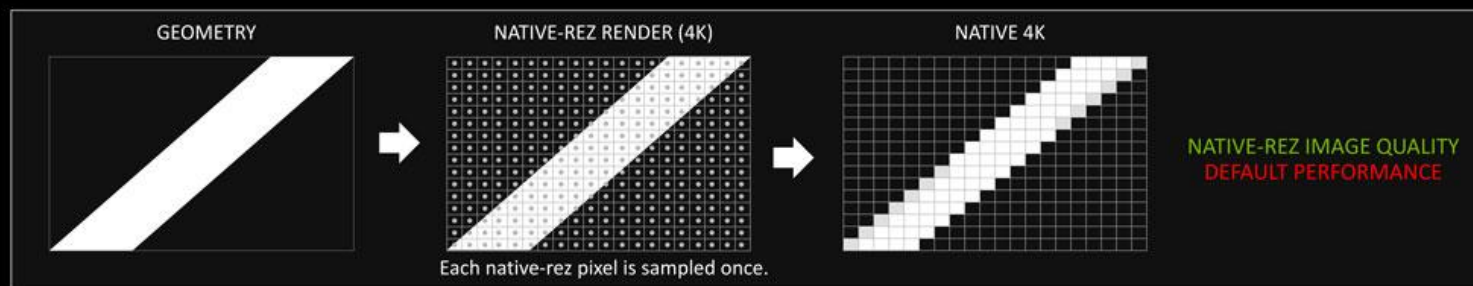
Ray Tracing



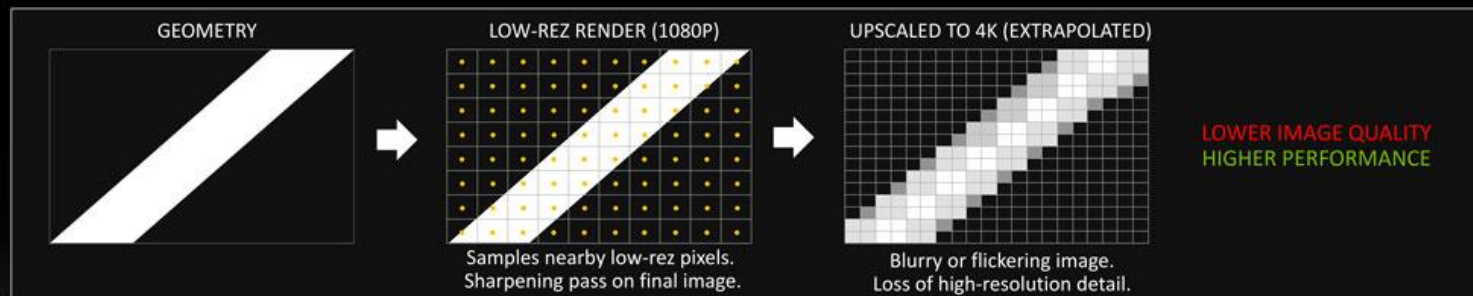
Deep Learning Super Sampling

NVIDIA DLSS VS. TRADITIONAL UPSCALING

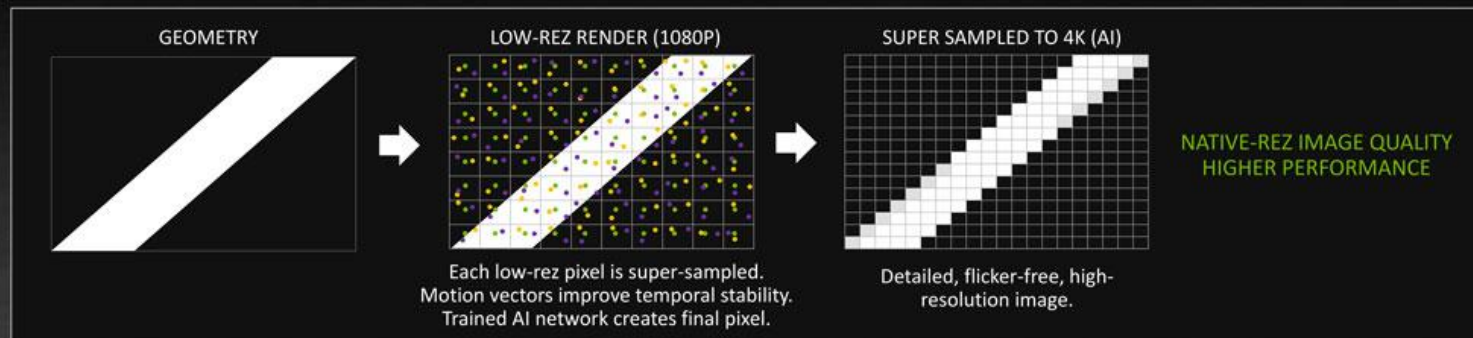
NATIVE



UPSCALE & SHARPEN



DLSS



Deep Learning Super Sampling

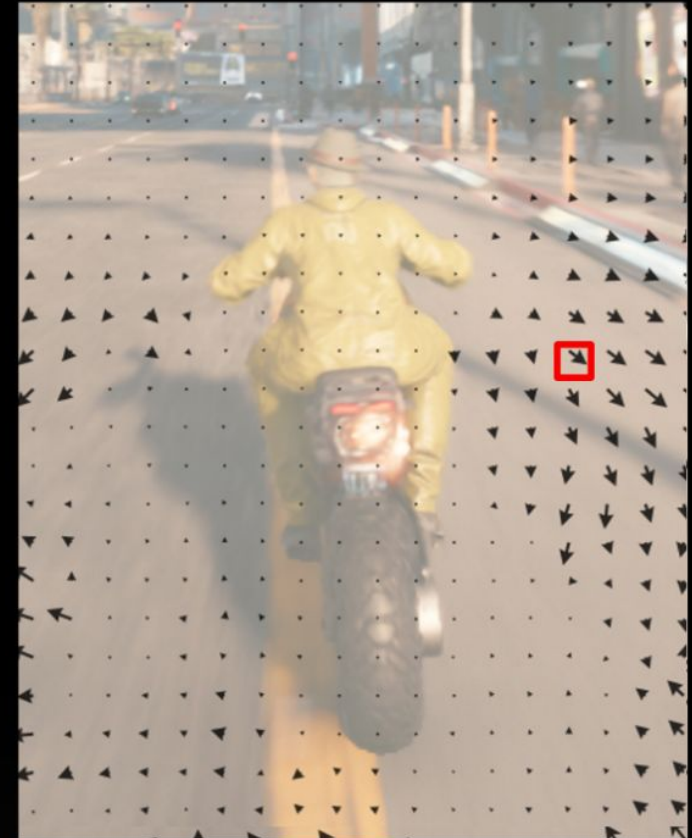
For each pixel region in frame 1



Find matching pixel region in frame 2

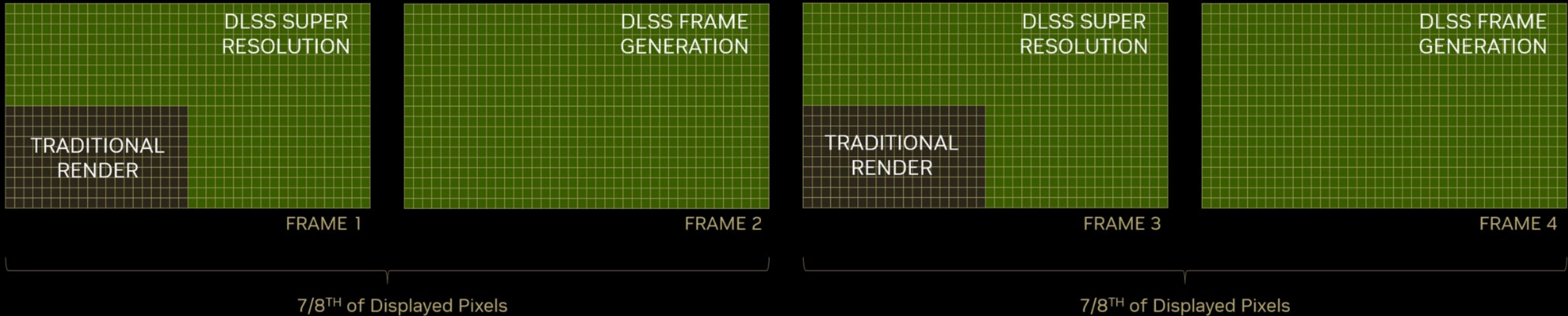


Calculate motion flow



Deep Learning Super Sampling

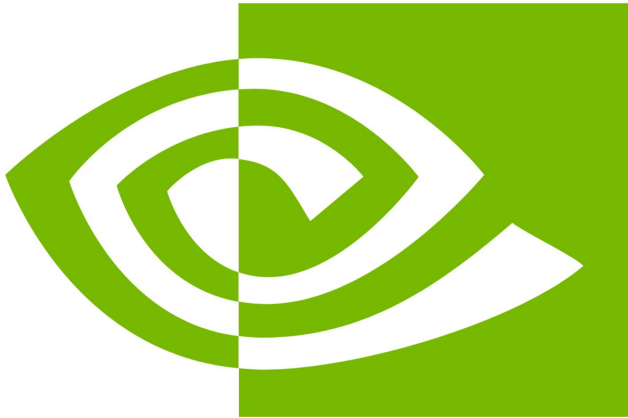
NVIDIA DLSS 3 RECONSTRUCTS 7/8TH OF DISPLAYED PIXELS



Deep Learning Super Sampling



NVIDIA Market Cap: \$3.42T



>

United Kingdom GDP: \$3.34T



INNOVATION FEST

- **Sign-up Deadline: February 19, 2025**
- **Challenges Released: February 21, 2025**
- **Final Competition: February 22, 2025**
- **This is a solo competition—no teams!**
- **Ensure you're present for the final round to compete for the top spot!**



Protecting National Security: Cyber Insights

DATE TBD | 5 - 6:15PM | ROOM TBD

- Learn about emerging cybersecurity challenges faced by the U.S. government
- See how the government's policies, regulations, and orders drive cybersecurity in all industries
- Gain insights on how to build a cybersecurity career in government

Pizza will be provided!



**Presented by Andy Seely,
Global IT Leader, IEEE CS VP of
Membership & Geographic
Activities**

